PDF AND CDF

PDF to CDF and back

Two representations of one distribution, linked by integration and differentiation.

PDF

$f_X(x)$

- Exponential: $\lambda e^{-\lambda x}$ for $x \geq 0$
- Uniform on $[0, 10]$: $\frac{1}{10}$

INTEGRATE →

CDF

$F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$

- Exponential: $1 - e^{-\lambda x}$
- Uniform: $\frac{x}{10}$ on $[0, 10]$
- Continuous and non-decreasing

DIFFERENTIATE →

Back to the PDF

$f_X(x) = \frac{d}{dx} F_X(x)$

- The Fundamental Theorem of Calculus applied to probability
- Steeper CDF, higher PDF; flat CDF, zero density